

IBM DATA ANALYST CAPSTONE PROJECT TECHNOLOGY TRENDS ANALYSIS

Prepared by:
ANNAH MOKOO

Date:
23 July 2026

© IBM Corporation. All rights reserved.

OUTLINE

1. Executive Summary
2. Introduction
3. Methodology
4. Results
5. Programming Language Trends
6. Programming Language Trends – Findings and Implications
7. Database Trends
8. Database Trends – Findings and Implications
9. Job Postings
10. Popular Languages
11. Dashboard Tab1 – Current Technology Usage Dashboard
12. Dashboard Tab2 – Future Technology Trends Dashboard
13. Dashboard Tab3 – Demographics Dashboard
14. Discussion
15. Overall Findings and Implications
16. Conclusion
17. Appendix

EXECUTIVE SUMMARY



- This project analyzes data to identify current and emerging technology trends in the software industry.
- This analysis focuses on understanding demographics, technology usage, future technology preferences, and employment demand through visualizations and dashboards.
 - Cleaned and analyzed data
 - Created visualizations and interactive dashboards
 - Collected data through APIs and Web scraping
- Key findings- Python, JavaScript, SQL, Cloud platforms, and modern databases are the most in-demand and preferred technologies.
- Insights- The dashboards highlight developer demographics, current technology usage, future technology trends and job market demand.
- Impact- The findings support career planning, workforce development and informed technology adoption decisions.

INTRODUCTION



Purpose:

- To analyze Stack Overflow Developer Survey data together with job market data to identify current and emerging technology trends.

Target Audience:

- Developers, employers, educators, students and organizations interested in technology workforce trends.

Project Scope:

- The analysis focuses on developer demographics, current technology usage, future technology preferences, and employment trends.

Value:

- The findings support informed decisions on career development, recruitment, technology adoption, workforce planning, and curriculum design.

METHODOLOGY

Data Sources

- Stack Overflow Developer Survey
- job posting APIs
- web scraping sources.

Data Collection

- API
- Web Scraping
- Survey Dataset

Data Wrangling

- Cleaning
- Filtering
- Transformation

Tools

- Python
- SQL
- IBM Cognos



RESULTS

The analysis produced several key findings:

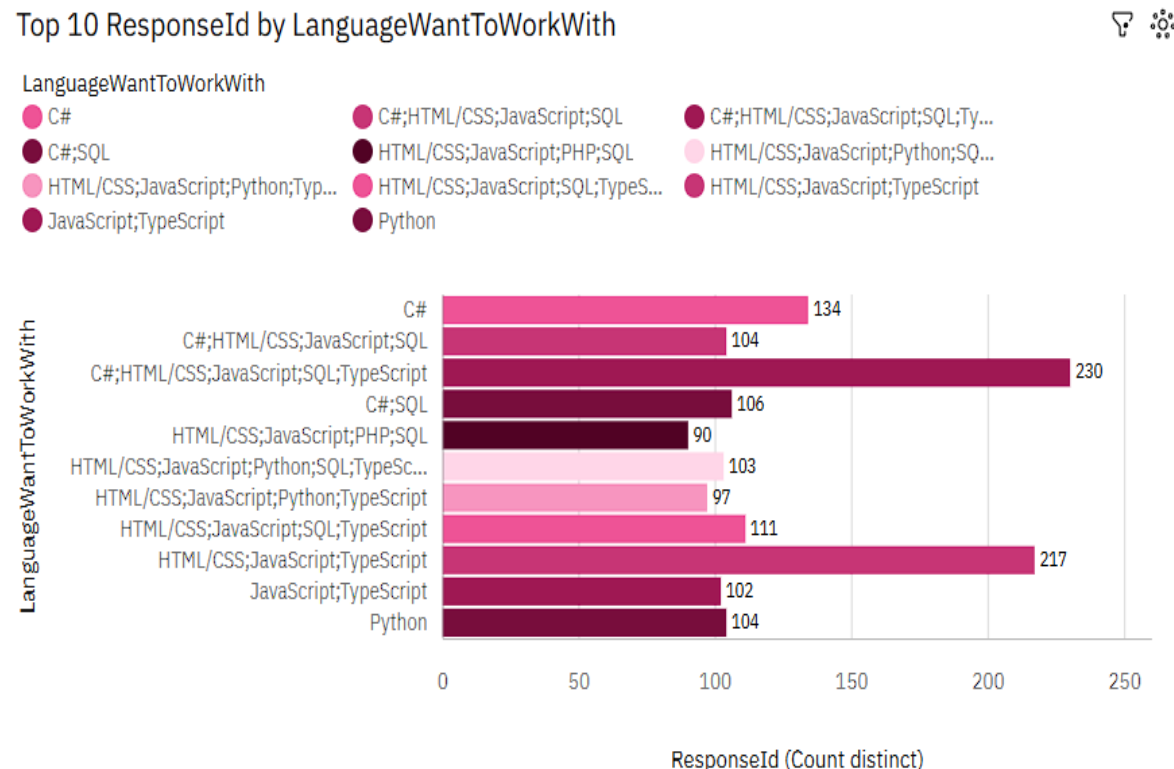
- Python, JavaScript, and SQL were among the most commonly used programming languages.
- PostgreSQL and MongoDB showed strong growth as preferred future databases.
- Cloud platforms such as AWS and Azure remain widely used.
- Developers expressed growing interest in modern web frameworks including React and Node.js.
- Most survey respondents were between 25 and 34 years old and held at least bachelor's degree.
- The most widely used and most in-demand programming languages, databases, platforms, and web frameworks while highlighting future technology trends.

PROGRAMMING LANGUAGE TRENDS

Figure 1.
Bar Chart – Top 10 Programming Languages(Current Year)



Figure 2.
Bar Chart – Top 10 Programming Languages(Next Year)



Key Observation

- Python and JavaScript remain the most widely used and highly desired programming languages.

PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS



Implications

Developers should prioritize learning Python, JavaScript, SQL and cloud technologies. Organizations should prioritize these languages when planning recruitment , software development, and employee training. Educational institutions can update curricula to reflect current industry demands.

Findings

Python and JavaScript remain highly demanded programming languages. SQL continues to be an essential skill across industries. Cloud computing platforms continue to dominate the technology landscape.

Overall findings

Python, JavaScript, and SQL remain essential skills across industries.

DATABASE TRENDS

Figure 3.
Bar Chart – Database Trends(Current Year)

Top 10 DatabaseHaveWorkedWith



DatabaseHaveWorkedWith

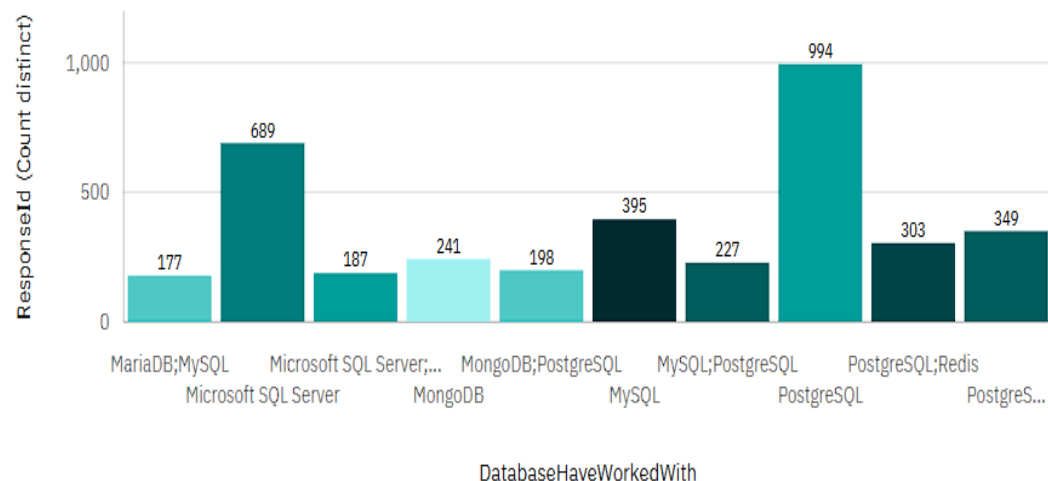
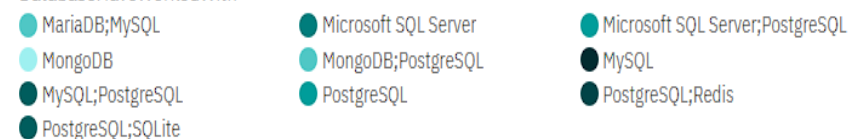
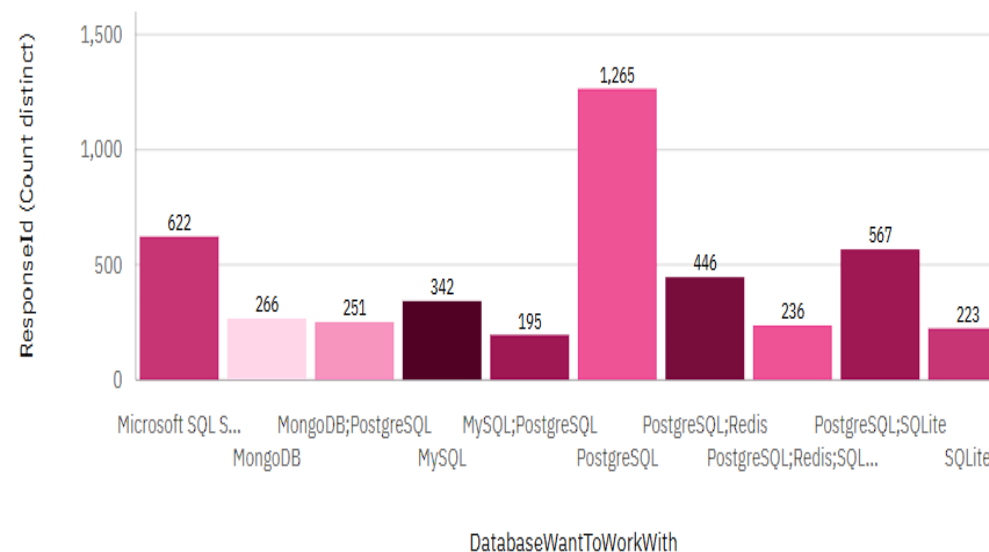
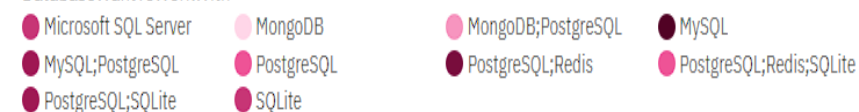


Figure 4.
Bar Chart – Database Trends(Next Year)

Top 10 ResponseId by DatabaseWantToWorkWith



DatabaseWantToWorkWith



Key observation

- PostgreSQL and MongoDB show increasing future demand.

DATABASE TRENDS - FINDINGS & IMPLICATIONS



Findings

PostgreSQL, MongoDB, and MySQL are among the most popular and preferred databases. Developers show increasing interest in modern, open-source databases for future projects. Traditional Relational databases remain important, while NoSQL databases continue to grow in adoption.

Overall Finding

Modern databases continue to gain popularity while relational databases remain important.

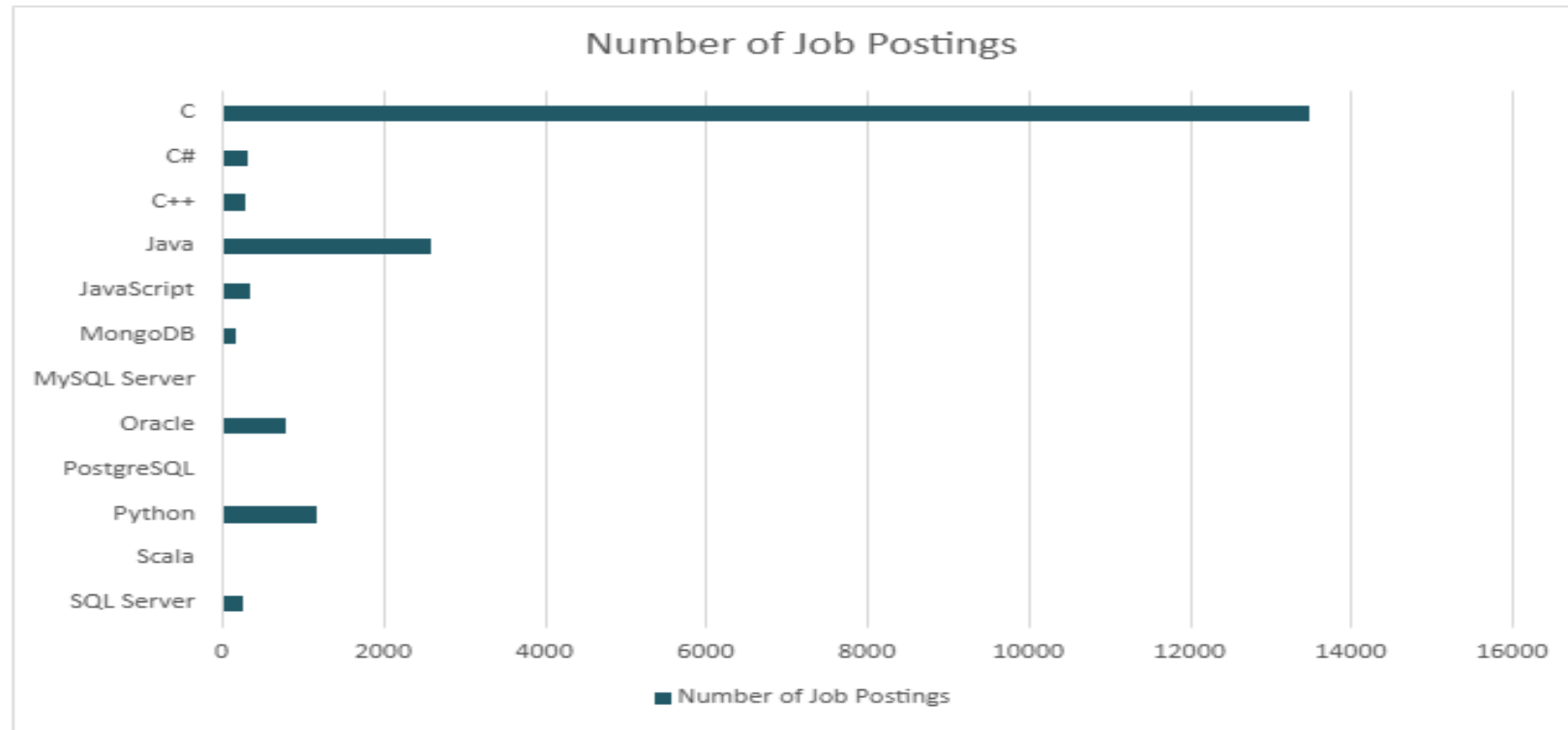
Implications

Developers should build expertise in both SQL and NoSQL databases to remain competitive. Organizations should consider adopting modern database technologies to improve scalability and performance. Training programs and educational institutions should include both relational and NoSQL database technologies to meet industry demand.

JOB POSTINGS

Figure 5.

Bar Chart – Job Postings



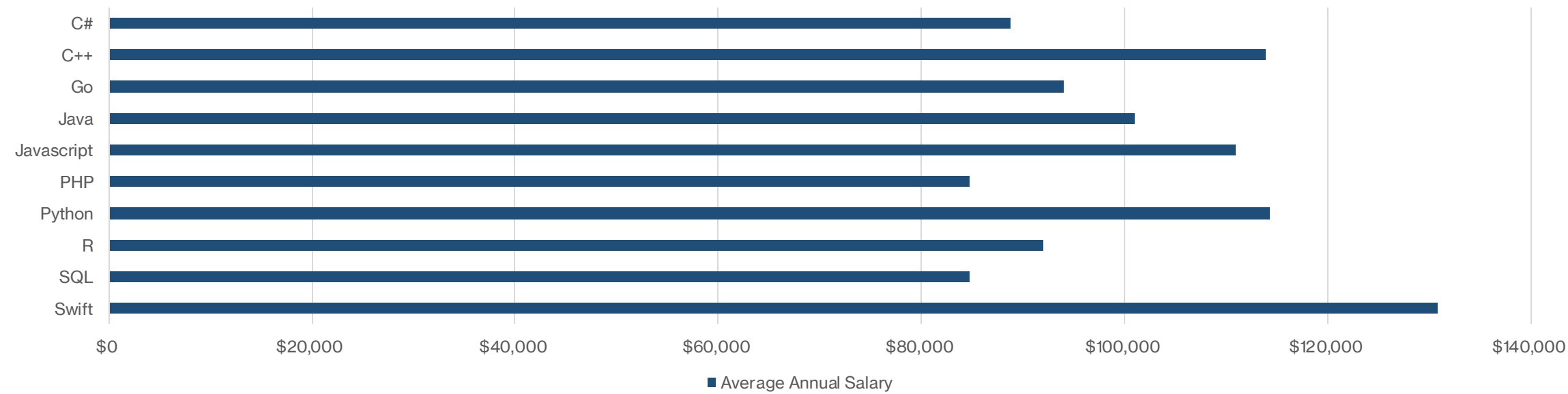
Observation

- Python has the highest number of job postings followed by JavaScript and SQL.

POPULAR LANGUAGES

Figure 6.
Bar Chart – Popular Languages

Average Annual Salary



Observation

- Python offers one of the highest average salaries among the selected programming languages.

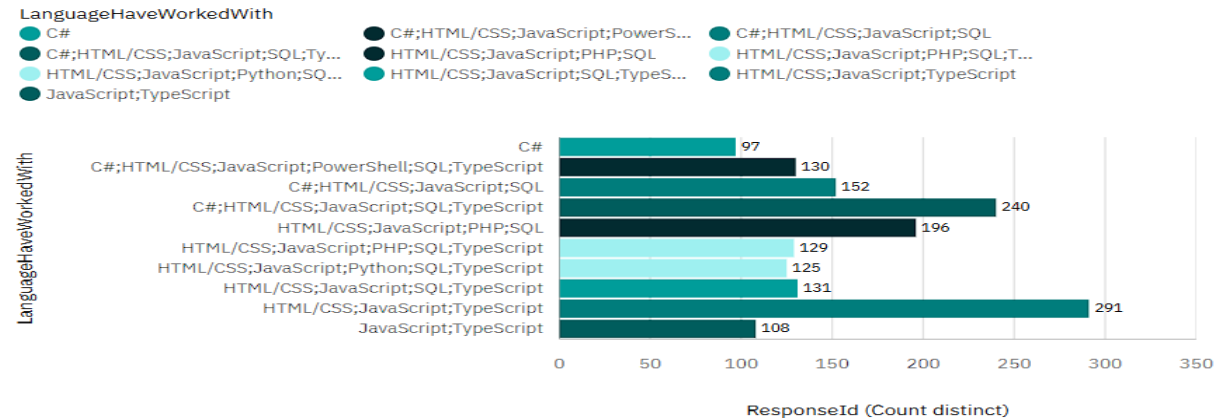
DASHBOARD

https://us1.ca.analytics.ibm.com/bi/?perspective=dashboard&pathRef=.my_folders%2FDemographics&action=view&mode=dashboard&subView=model0000019f7f3c4741_000000000&nav_filter=true

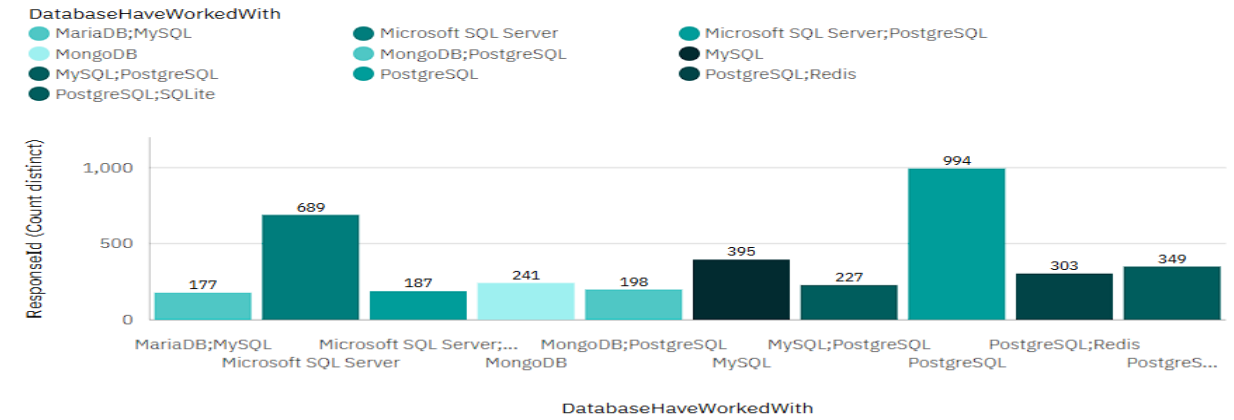
DASHBOARD TAB1- CURRENT TECHNOLOGY USAGE

- ✓ Top 10 Programming Languages
- ✓ Top 10 Databases
- ✓ Top 10 Platforms
- ✓ Top 10 Web Frameworks

Top 10 LanguageHaveWorkedWith



Top 10 DatabaseHaveWorkedWith



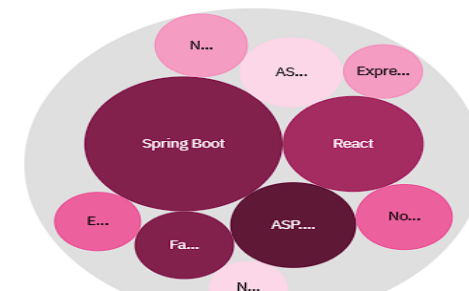
Top 10 PlatformHaveWorkedWith



Top 10 WebframeHaveWorkedWith hierarchy



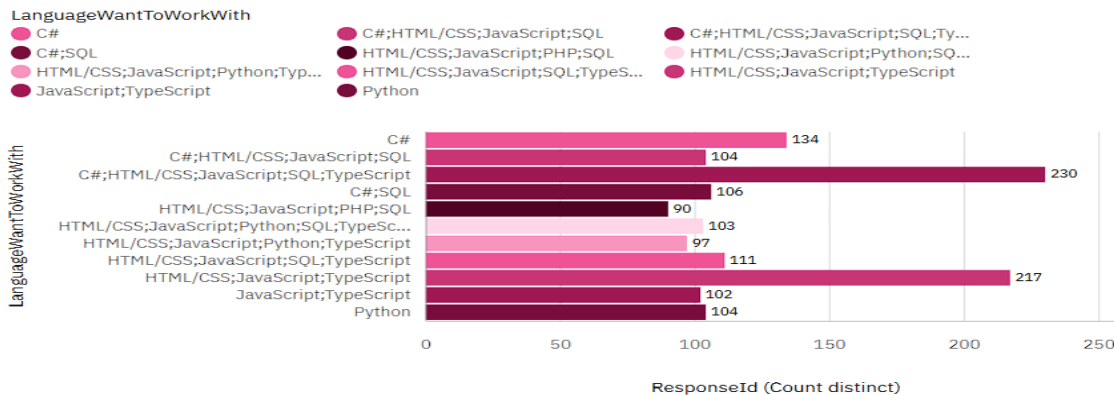
Amazon Web Services (AWS);Digital Ocean;Firebase;V



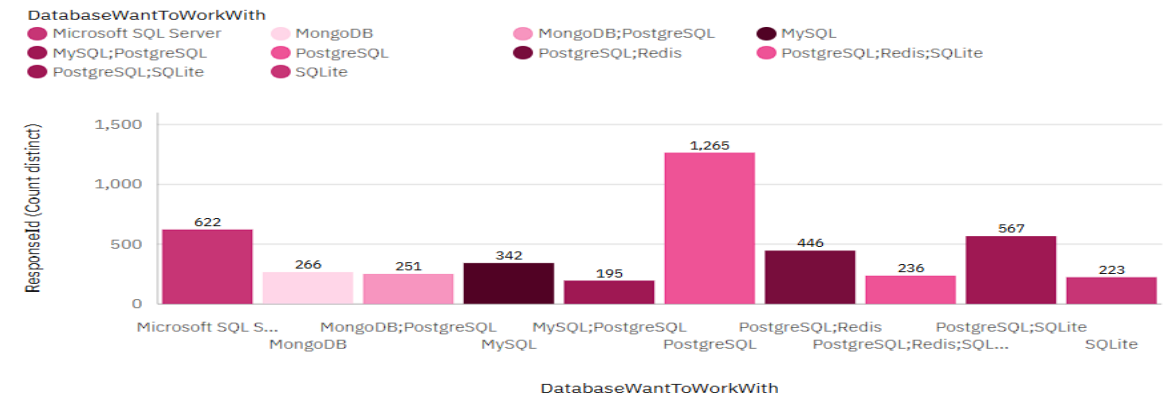
DASHBOARD TAB2- FUTURE TECHNOLOGY TRENDS

- ✓ Future Programming Languages
- ✓ Future Databases
- ✓ Future Platforms
- ✓ Future Web Frameworks

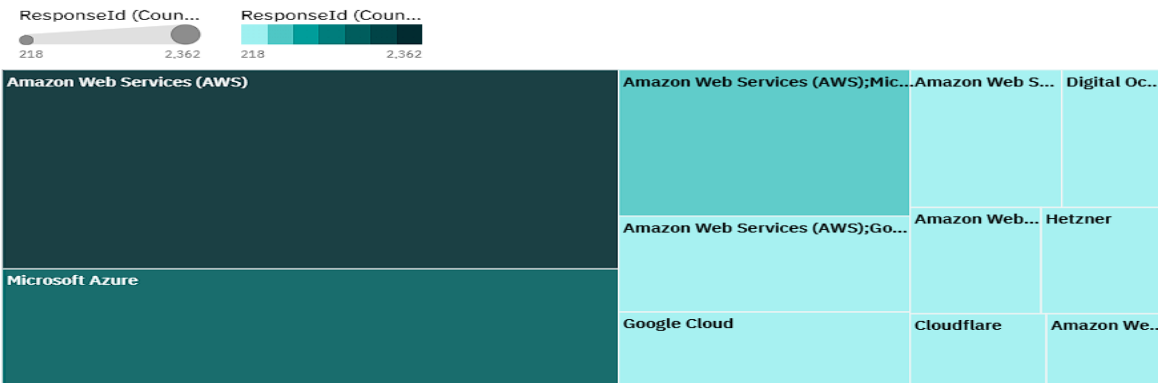
Top 10 ResponseId by LanguageWantToWorkWith



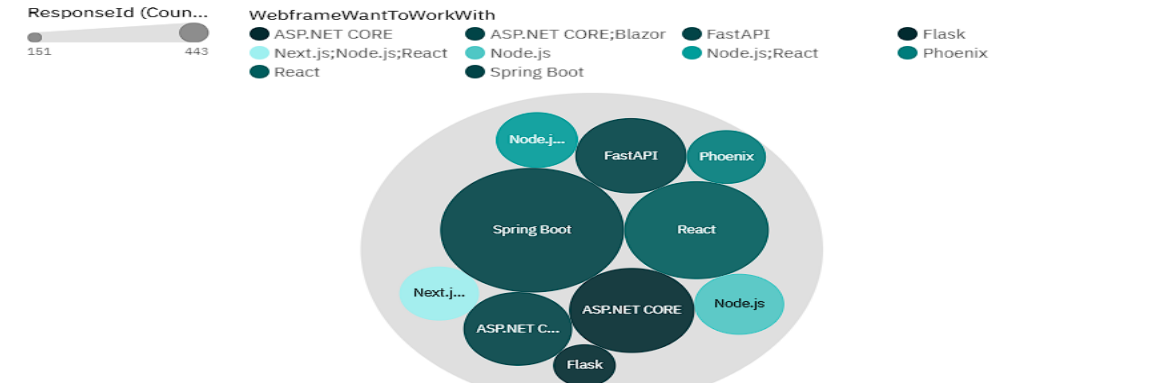
Top 10 ResponseId by DatabaseWantToWorkWith



Top 10 PlatformWantToWorkWith hierarchy



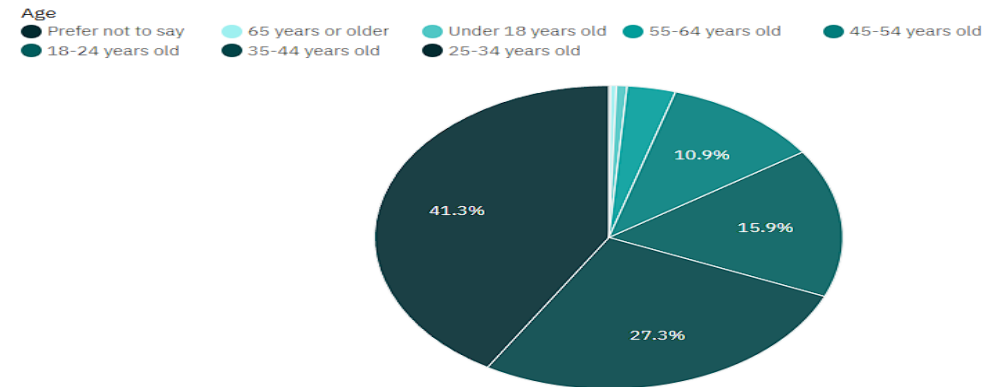
Top 10 WebframeWantToWorkWith hierarchy



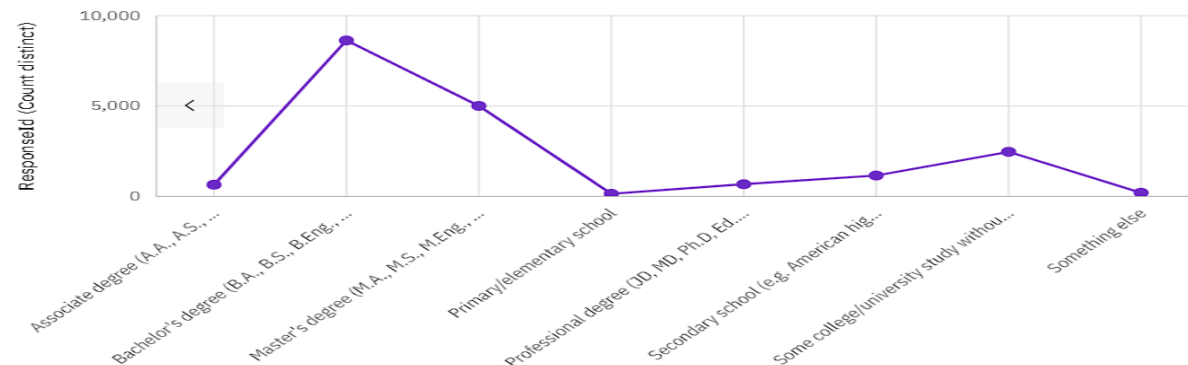
DASHBOARD TAB3- DEMOGRAPHICS DASHBOARD

- ✓ Respondent Age
- ✓ Respondent Country
- ✓ Educational Level
- ✓ Age by Educational Level

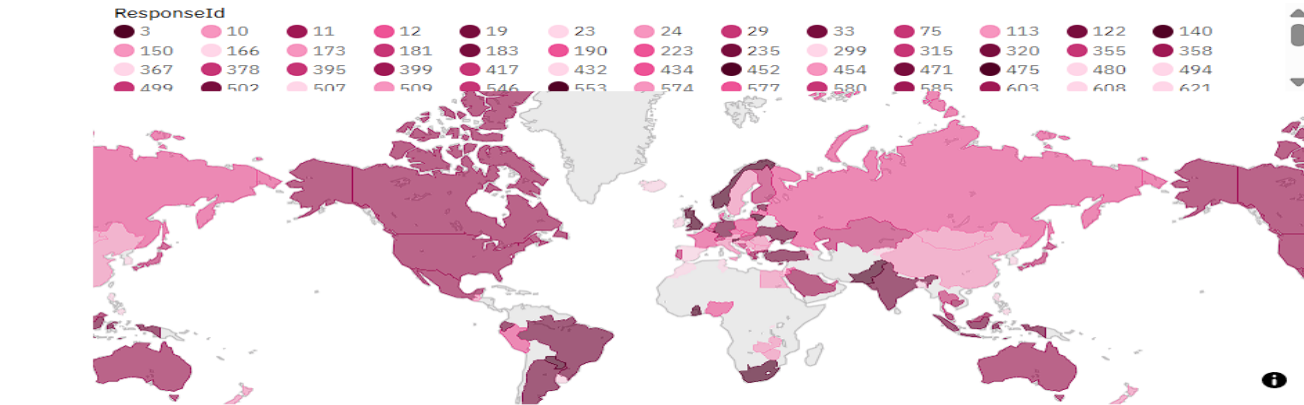
ResponseId by Age



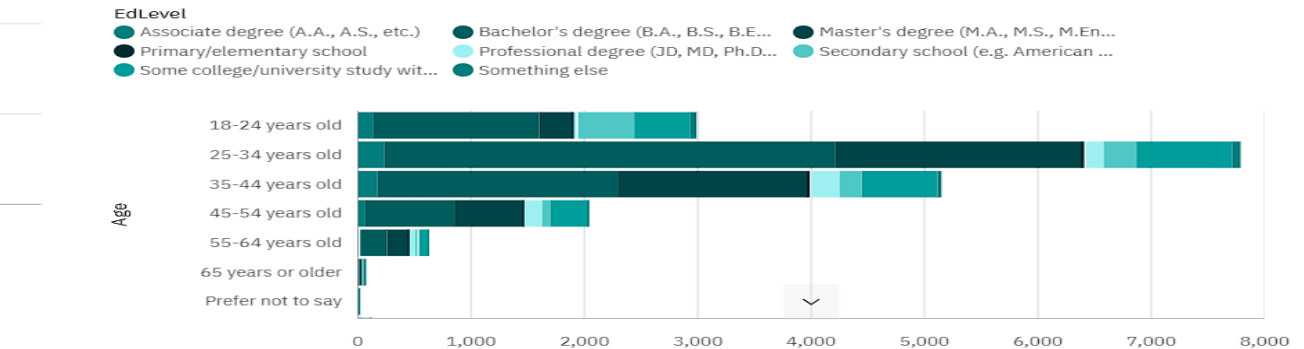
ResponseId by EdLevel



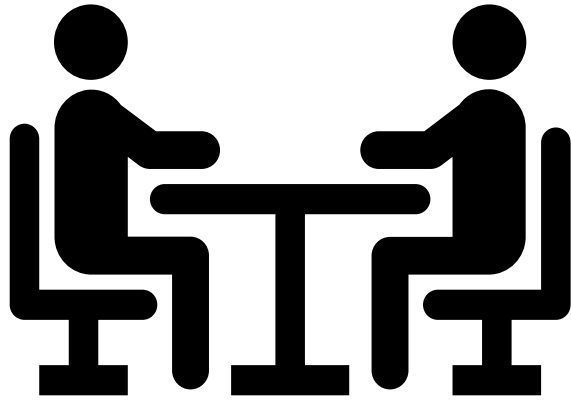
ResponseId by Country regions



ResponseId by Age colored by EdLevel



DISCUSSION



The analysis demonstrates that while established technologies such as JavaScript, Python, SQL and cloud platforms continue to dominate the industry, developers increasingly prefer modern technologies like PostgreSQL, MongoDB, React, and cloud-native platforms. The comparison between current usage and future preferences suggests a gradual shift toward newer technologies while maintaining demand for widely adopted programming languages.

Overall, the analysis shows that established technologies remain relevant while modern technologies continue to gain adoption.

OVERALL FINDINGS & IMPLICATIONS

Findings

Python, JavaScript, SQL, and cloud technologies remain the most widely used and in-demand skills.

Developers are increasingly interested in modern databases, web frameworks, and cloud platforms for future work.

The analysis highlights changing technology preferences, reflecting the industry's shift toward modern and scalable technologies.

Implications

Developers should continuously update their skills to stay aligned with current and emerging technology trends.

Organizations can use these insights to guide hiring, workforce planning , and technology investment decisions.

Educational institutions should align curricula with industry demands to better prepare learners for technology careers.

Overall Conclusion

Technology trends indicate continuous growth in cloud computing, modern databases, and python-based development.

CONCLUSION



- Successfully analyzed developer survey and job market data to identify current and future technology trends.
- Developed visualizations and dashboards that clearly present technology adoption, developer preferences, and employment demand.
- Identified key technologies and skills that are in demand and expected to remain relevant in the future.
- The insights can support career planning, organizational decision-making, and curriculum development for the technology industry.

Project Outcome

The project successfully met its objective of identifying current and future technology trends using survey and job market data

APPENDIX

https://us1.ca.analytics.ibm.com/bi/?perspective=dashboard&pathRef=.my_folders%2FDemo%2Fgraphics&action=view&mode=dashboard&subView=model0000019f7f3c4741_000000000&nav_filter=true

Software Used

- IBM Cognos Dashboard Embedded
- Python
- SQL
- Microsoft PowerPoint
- Jupyter Notebook

Data sources

- Stack Overflow Developer Survey
- Job posting API dataset
- Programming languages salary dataset

Tools & Technologies

- Python(Pandas, Matplotlib, BeautifulSoup, Requests)
- SQL(SQLite)
- IBM Cognos Dashboard Embedded
- Jupyter Notebook
- Microsoft PowerPoint

Analysis Performed

- Data cleaning and preprocessing
- API data collection
- Web scraping
- Exploratory Data Analysis
- Data visualization

References

- Stack Overflow Developer Survey
- IBM Skills Network Labs